Sci-Fi Warp Drive & Wormholes

Overview **2**
Asset setup **3**
URP 3
HDRP 3
Warp Shader configuration **4**
Warp Effect (VFX Graph) **4**
Warp Controller **7**
Starting the scene and activating the effect **7**
Camera Shaker: **8**
Integration in different platforms **8**

Overview

This versatile asset empowers you to shape the very fabric of space and time in your sci-fi game, offering a rich palette of customizable layers for a truly unique and immersive experience.

Key features:

Layered and fully customizable Warp Drive Effect
Craft mesmerizing wormhole experiences with highly customizable portal effects. Adjust colors, sizes, and rotation speeds to suit the unique aesthetics of your game. This asset includes a VFX graph effect for the warp drive and a warp shader for more customizable warp & wormhole effects. Use them individually or combine them creatively to craft distinctive visual effects tailored to your game.

Color and Intensity Adjustments
Tailor the visuals to perfection with the ability to adjust colors, effect intensity, and color intensity. Use your own textures in combination with the warp shader to create unique and individual effects. Whether you seek a subtle journey through the stars or a vibrant warp experience, the power is in your hands.

Dynamic Triggering with WarpController Script
Seamlessly trigger warp drive effects during gameplay events with the included WarpController script. Immerse players in the excitement of interstellar travel at the touch of a button.

Camera Shaker
Camera Shaker script integrated in the WarpController settings, allowing you to additionally trigger camera shaking based on your warp configurations, which makes your warp drive effect even more realistic.

Intuitive Integration
Seamlessly integrate the warp drive and wormhole effects into your Unity project with user-friendly scripts and documentation. No need to be a coding expert – our asset is designed for easy implementation.

Asset setup

URP

The asset is already developed in URP, so the asset should work out of the box.
Just please make sure that you have Shadergraph and Visual Effects Graph installed.

HDRP

If you want to open the asset in an HDRP project, you can find a HDRP folder with a unitypackage with all the HDRP updated. Just import everything from the default package (URP version). Then double click on the HDRP unitypackage and import it.
This will replace all the URP materials and shaders graphs with the HDRP versions.

After that you are ready to use the package.

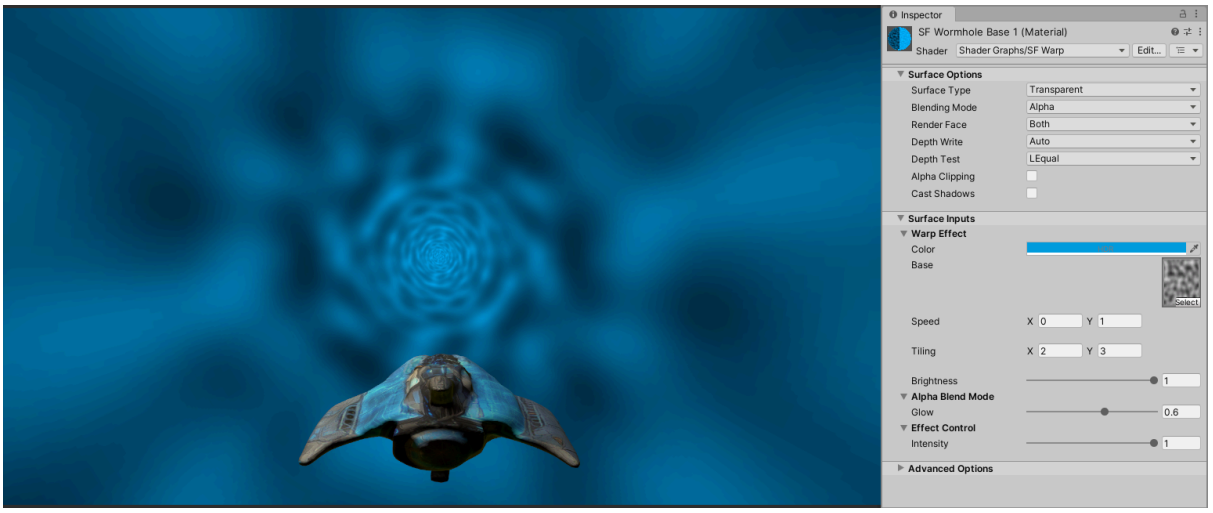
Warp Shader configuration

Use the *SF Warp Shader* for creating various warp and wormhole effects.
In the demo scene most of the effects contain 2-3 effects made with this shadergraph to add more detail to the wormholes .You can find the shadergraph under the 'SF Warp Drive & Wormholes → ShaderGraphs' folder.

Configuration:

Color	Specify the main color of your effect
Base	The warp shader is basically scrolling a given texture with a given speed and adding some additional effects. Change the Texture to create unique effects. The textures used in this project can be found in 'SF Warp Drive & Wormholes → Textures
Speed	Define the scroll speed in x and y axis
Tiling	Define how often the given texture should be duplicated in x and y axis
Brightness	Specify the brightness of the effect.
Glow (Alpha blend)	**Note: This is only working with Blending Mode: Alpha.** This mode is currently used for the *SF Wormhole Base* materials to increase the shining from the center of the wormhole
Intensity	Exposed value that is used to activate and deactivate the effects in runtime. Value of zero means there is no effect. With a value of 1 the effect will be shown at its full strength.

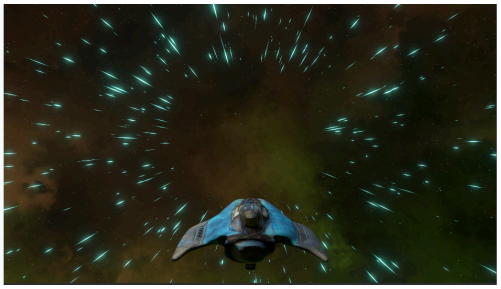
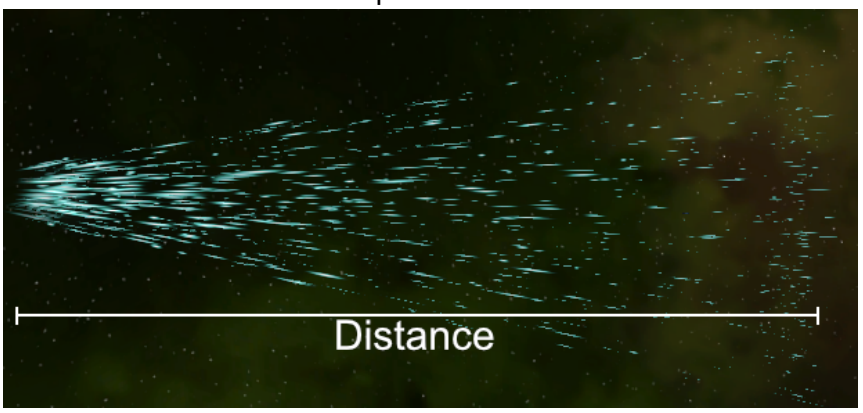
Example:



Warp Effect (VFX Graph)

This VFX effect creates a stunning warp drive effect, giving the user an illusion of traveling through space.

Configuration:

Base	Effect texture. This effect is made using the default particle from unity, but can be exchanged by any texture you like.
Color	Color of the particles
Max rate	Max spawn rate. Set this to a higher value if you want more particles to spawn at the same time.
Max Attraction Speed	Particles are spawned within the spawn radius and will fly towards you until the attraction distance is reached. Increase the speed value to make the particles fly faster towards you.
Spawn Radius	Radius in which the particles will spawn. Increase the value to make the center less dense. This example shows how the effect looks with a spawn radius of 2 instead of the default value of 0.2 
Attraction Distance	The max distance where the particles are attracted towards. 
Intensity	Exposed value that is used to activate and deactivate the effects in runtime. Value of zero means there is no effect. With a value of 1 the effect will be shown at its full strength.

Example:

Transform
Visual Effect
General
Asset Template
Warp
Edit
Random Seed
0
Reseed
Reseed on play
Initial Event Name
OnPlay
Renderer
Instancing
Properties
Show Property Gizmos
General
Base
Default-Particle
Color
HDR
Max Rate
800
Max Attraction Speed
5
Sizing
Spawn Radius
0.2
Attraction Distance
6
Effect Control
Intensity
1
Add Component

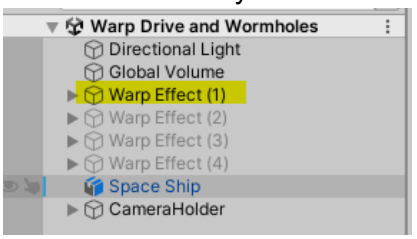
Warp Controller

There are 4 warp effects added in the demo scene. Every effect has a Warp Controller script attached to it. The controller will handle activating and configuring the warp and wormhole effects for you.
Configuration:

Warp VFX	Contains the reference of the Warp VFX Graph effect
Wormhole Effects	Contains a list of all other *SF Warp shader* effects.
Wormhole Delay	Since the warp VFX needs a bit of time to start, we can specify a delay for all the additional effects that are using the Warp Shader
Start Duration	Specify the duration in seconds until the effect reaches the max intensity
Hold Duration	Specify the duration in seconds of how long the effect should stay at the max intensity
Stop Duration	Specify the cooldown time in seconds. That's the time that the effect needs to get from the max intensity to a complete stop
Magnitude	There is an additional integrated Camera Shaker script attached to the Main Camera. You only need to specify the strength of the camera shake, all the rest will be handled automatically. The shake will also build up according to the start duration and cooldown according to the cooldown duration

Starting the scene and activating the effect

Just make sure that one of the Warp Effect gameobjects is activated in the hierarchy.



Start the scene and simply **press space** to activate the effect.
You can disable and enable the warp effects in the hierarchy without needing to stop your scene.
All the enabled effects will start when pressing space.

You can also change the warp controller configuration during runtime to play around with the effect and find a configuration that suits you the best. (But please make sure that you copy the values before you stop the game, otherwise your changes will be gone!)

Camera Shaker:

The camera shaker is added as an additional feature to better showcase the warp effect and make it feel more “realistic”. Since the shaker randomly changes the x and y transform positions of the attached gameObject, it could potentially interfere with your own camera scripts.Depending on how your camera controller works it could also simply negate the shake effects. So keep in mind that this script may or may not work with your controller.
But feel free to look at the code and directly integrate it in your camera controller.
Also if you are stuck and need help, feel free to contact me.

Note:
If you use the shaker, please make sure the CameraShaker script is attached to your camera.

If you don’t want to use the Camera Shaker, simply remove the *Camera.Instance.Shake()* part from the *StartEffect()* method in the WarpController.

Integration in different platforms

The added warp controller per default listens for the space key of the keyboard. The effect itself should be platform independent, so the only thing that could bug you is the input handling.

```
void Update() {
    if (Input.GetKey(KeyCode.Space)) {
        StartEffect();
    }
}
```

To make it working on VR, mobile or other platforms you have 2 options:

1. Change the Input check from *Input.GetKey(KeyCode.Space)* to a platform specific input
2. Or Simply delete the *Update()* function and directly call *StartEffect()* when clicking e.g. on a button that should trigger the effect.

Hope you enjoy this asset!
For any questions/ complaints/suggestions contact me under: ***gamedevibk@gmail.com***